# Cloud Computing for Machine Learning

*Introduction to Cloud Computing*

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# Course Walkthrough

## Weekly Topics

## Assessment

- 5 Labs – 40 %
- 3 Projects – 60%

## Tools & Frameworks

- Various cloud platforms!

**Teams Space**

Join our class on Teams with this code: **06sfyj6**

All online sessions and important announcements will be on Teams!

# Agenda

- Introduction to Cloud Computing
- Why Cloud Computing for Machine Learning?
- Core Components of Cloud Computing
- Popular Cloud Platforms for Machine Learning
- Types of Cloud Services
- Challenges and Considerations
- Q&A

# What is Cloud Computing?

Cloud computing is the delivery of computing services over the internet, offering faster innovation, flexible resources, and economies of scale.

The main characteristics of cloud computing are: On-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service.

# Why Cloud Computing?

**Scalability:** Ability to handle large datasets and computational needs.

**Flexibility:** Access to a wide range of tools and technologies.

**Cost-effectiveness:** Pay-as-you-go models reduce upfront costs.

**Collaboration:** Easier sharing and collaboration on projects.

# Core Components of Cloud Computing

**Infrastructure as a Service (IaaS):**
Basic building blocks for cloud IT.

**Platform as a Service (PaaS):**
Allows developers to build applications.

**Software as a Service (SaaS):**
Software available via third-party over the internet.

**Data Storage and Analytics:**
Storing and analyzing large datasets.

# Popular Cloud Platforms for Machine Learning

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)
- IBM Cloud

# Types of Cloud Services

**Public Cloud:** Services offered over the public Internet.

**Private Cloud:** Exclusive services for a single organization.

**Hybrid Cloud:** Combination of both public and private cloud services.

**Community Cloud:** Shared by multiple organizations with similar needs.

# Challenges and Considerations

**Data Privacy and Security:** Protecting sensitive information.

**Compliance and Legal Issues:** Adhering to regulations.

**Cost Management:** Avoiding unexpected expenses.

**Skill Requirements:** Need for specialized knowledge.

# Summary and Key Takeaways

**Cloud Computing Benefits:** Scalability, flexibility, and cost-effectiveness in machine learning.

**Platform Selection:** Choosing the right platform based on specific needs.

**Implementation Strategy:** Following best practices for successful deployment.

**Future Trends:** Continuous evolution in cloud technology and machine learning integration.

# Q&A